Weak Compactness in the Column Martingale Hardy Space

Let $(\mathcal{M},\tau)$ be a finite von Neumann algebra equipped with a faithful normal tracial state, so that $\tau(1)=1$. Let $(\mathcal{M}_n)_{n\ge 1}$ be an increasing filtration with trace-preserving conditional expectations $\mathcal{E}_n$. For a martingale $x=(x_n)_{n\ge 1}$, put

$$dx_n = x_n - x_{n-1}, \qquad x_0 = 0, \qquad S_c(x) = \left(\sum_{n\ge 1} dx_n^* dx_n\right)^{1/2}.$$

The column Hardy space $H_1^c = H_1^c(\mathcal{M})$ is the completion of finite martingales under the norm

$$\|x\|_{H_1^c} = \|S_c(x)\|_1.$$

A family $\mathcal{F}\subset L_1(\mathcal{M})_+$ is called *uniformly integrable* if it is norm bounded and

$$\lim_{R\to\infty}\sup_{h\in\mathcal{F}} \tau\Big(h\,\mathbf{1}_{(R,\infty)}(h)\Big) = 0.$$

For a finite trace, the displayed condition already implies norm boundedness. All weak topologies below are Banach-space weak topologies.

Theorem 1. *A subset $K\subset H_1^c$ is relatively weakly compact if and only if*

$$S_c(K) = \{S_c(x) : x\in K\}$$

is uniformly integrable in $L_1(\mathcal{M})$.

We first record the consequence of Akemann's weak compactness criterion that will be used in the proof. The criterion says that a bounded subset $A\subset L_1(\mathcal{N},\nu)$ is relatively weakly compact if and only if

$$\lim_{m\to\infty}\sup_{a\in A}\|q_m a q_m\|_1 = 0$$

for every decreasing sequence of projections $q_m\downarrow 0$ in $\mathcal{N}$. Here $(\mathcal{N},\nu)$ may be semifinite; see [1, Definition 2.4 and the subsequent remarks].

Lemma 1. *Let $(\mathcal{N},\nu)$ be a semifinite von Neumann algebra with a faithful normal semifinite trace. If $A\subset L_1(\mathcal{N},\nu)$ is relatively weakly compact, then*

$$\lim_{\delta\downarrow 0}\sup\big\{\|qaq\|_1 : a\in A,\ q\in\mathcal{N}\ \text{a projection},\ \nu(q)\le\delta\big\} = 0. \tag{1}$$

Proof. If (1) fails, there exist $\varepsilon>0$, $a_j\in A$, and projections $r_j\in\mathcal{N}$ such that

$$\nu(r_j)\le 2^{-j}, \qquad \|r_j a_j r_j\|_1\ge\varepsilon \quad (j\ge 1).$$

Define $q_m = \bigvee_{j\ge m} r_j$. Then q_m decreases and

$$\nu(q_m) \le \sum_{j\ge m} 2^{-j} \longrightarrow 0.$$

Faithfulness and normality of ν imply $q_m \downarrow 0$. Since $r_m \le q_m$,

$$\|q_m a_m q_m\|_1 \ge \|r_m(q_m a_m q_m) r_m\|_1 = \|r_m a_m r_m\|_1 \ge \varepsilon.$$

This contradicts Akemann's criterion. □

For completeness, we also recall a standard weak compactness approximation principle. If E is a Banach space and a bounded set $A \subset E$ satisfies

$$A \subset W_\varepsilon + \varepsilon B_E \tag{2}$$

for every $\varepsilon > 0$, where $W_\varepsilon \subset E$ is weakly compact, then A is relatively weakly compact. Indeed, identify E with its canonical image in E^{**} and let C be the weak-star closure of A in E^{**}. The set $W_\varepsilon + \varepsilon B_{E^{**}}$ is weak-star compact, so (2) implies

$$C \subset W_\varepsilon + \varepsilon B_{E^{**}}.$$

Consequently $\operatorname{dist}(z, E) \le \varepsilon$ for every $z \in C$ and every $\varepsilon > 0$. Since E is norm closed in E^{**}, we have $C \subset E$. Thus C is weakly compact in E.

Proof of Theorem 1. We may assume $K \ne \varnothing$. Set

$$\mathcal{N} = \mathcal{M}\bar{\otimes}B(\ell_2), \qquad \nu = \tau \otimes \operatorname{Tr}, \qquad p = 1 \otimes e_{11}, \qquad E = L_1(\mathcal{N}, \nu)p,$$

where $(e_{ij})_{i,j\ge1}$ are the standard matrix units. In particular, $\nu(p) = 1$. The column map

$$J : H_1^c \longrightarrow E, \qquad Jx = \sum_{n\ge1} dx_n \otimes e_{n1},$$

is an isometric linear embedding. More precisely,

$$|Jx| = S_c(x) \otimes e_{11}, \qquad \|Jx\|_1 = \|x\|_{H_1^c}. \tag{3}$$

The map is first defined on finite martingales and then extended by completion. Its image is a norm-closed, hence weakly closed, subspace of E. Therefore K is relatively weakly compact in H_1^c if and only if $A = J(K)$ is relatively weakly compact in E, or equivalently in $L_1(\mathcal{N}, \nu)$.

Necessity. Suppose that K is relatively weakly compact. Then A is relatively weakly compact and

$$C_0 = \sup_{a\in A} \|a\|_1 < \infty.$$

Fix $a = u|a| \in A$ and $R > 0$. Let

$$e = \mathbf{1}_{(R,\infty)}(|a|), \qquad f = ueu^*, \qquad q = e \vee f.$$

Since e is dominated by the support of $|a|$, the operator f is a projection equivalent to e. Hence

$$\nu(q) \le \nu(e) + \nu(f) = 2\nu(e) \le \frac{2C_0}{R}. \tag{4}$$

Moreover, $fae = ae$, and thus

$$\nu\Big(|a|\,\mathbf{1}_{(R,\infty)}(|a|)\Big) = \|fae\|_1 = \|f(qaq)e\|_1 \le \|qaq\|_1. \tag{5}$$

Applying Lemma 1 to (4)–(5), we obtain

$$\lim_{R\to\infty} \sup_{a\in A} \nu\Big(|a|\,\mathbf{1}_{(R,\infty)}(|a|)\Big) = 0.$$

Together with (3), this proves that $S_c(K)$ is uniformly integrable.

Sufficiency. Suppose that $S_c(K)$ is uniformly integrable. For $a \in A$ and $R > 0$, define

$$a^{(R)} = a\,\mathbf{1}_{[0,R]}(|a|).$$

The truncated operator still has right support dominated by p. Since $\nu(p) = 1$,

$$\|a^{(R)}\|_2^2 = \nu\Big(|a|^2\mathbf{1}_{[0,R]}(|a|)\Big) \le R^2\nu(p) = R^2.$$

Also,

$$\begin{aligned}\eta(R) &:= \sup_{a\in A} \|a - a^{(R)}\|_1 \\ &= \sup_{a\in A} \nu\Big(|a|\,\mathbf{1}_{(R,\infty)}(|a|)\Big) \longrightarrow 0 \qquad (R\to\infty).\end{aligned} \tag{6}$$

The natural inclusion

$$L_2(\mathcal{N},\nu)p \longrightarrow E$$

is bounded: by the noncommutative Hölder inequality, for $b = bp$,

$$\|b\|_1 = \|bp\|_1 \le \|b\|_2\|p\|_2 = \|b\|_2.$$

Since $L_2(\mathcal{N},\nu)p$ is a Hilbert space, the image W_R of its closed ball of radius R is weakly compact in E. We have

$$A \subset W_R + \eta(R)B_E.$$

The approximation principle above and (6) show that A is relatively weakly compact in E. It follows from the closedness of $J(H_1^c)$ and the isometry of J that K is relatively weakly compact in H_1^c. □

Remark. The spectral truncations $a^{(R)}$ need not belong to $J(H_1^c)$. This causes no difficulty: weak compactness is first established in the ambient column space E, and then transferred to the closed subspace $J(H_1^c)$.

Remark (On the finite-trace hypothesis). The theorem is stated for a finite tracial probability space. It does not extend verbatim to arbitrary semifinite algebras.

For example, equip $B(\ell_2)$ with its usual trace and use the constant filtration $\mathcal{M}_n = B(\ell_2)$. For the single-step martingales with $x_n^{(k)} = e_{1k}$ for all $n \ge 1$, the set $\{x^{(k)} : k \ge 1\}$ is relatively weakly compact in H_1^c: the row space spanned by the e_{1k} is isometric to ℓ_2, and $e_{1k} \to 0$ weakly in S_1. However,

$$S_c(x^{(k)}) = e_{kk}.$$

This family is not uniformly integrable in the projection-decrease sense. Indeed, for $q_m = \sum_{k\ge m} e_{kk} \downarrow 0$,

$$\sup_{k\ge 1} \|q_m e_{kk} q_m\|_1 = 1 \qquad (m \ge 1).$$

Furthermore, spectral-tail decay alone does not characterize uniform integrability in this general semifinite setting.

References

[1] N. Randrianantoanina, *Sequences in non-commutative L^p-spaces*, J. Operator Theory **48** (2002), 255–272. Preprint: arXiv:math/0004144.